

CP020003 — Artificial Intelligence 2026

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Week 5— Homework Assignment

This week we move from the small, single-purchase-per-customer transactions dataset used in class into a real, order-level retail dataset with genuine repeat customers. That single difference matters: it means you can finally build a true RFM (Recency, Frequency, Monetary) feature set — the same technique we discussed in class but couldn't fully demonstrate on the smaller in-class dataset. Your job is to take the unsupervised learning workflow we practiced together — EDA, feature engineering, scaling, PCA, choosing K, clustering (K-Means, DBSCAN, or a deep-learning-based approach), validation, and business interpretation — and apply it end to end to this new dataset.

1. Dataset

Download the dataset directly from the course GitHub repository (no unzipping needed — it's a single CSV file):

```
https://github.com/kaopanboonyuen/CP020003\_ArtificialIntelligence\_2026s1/blob/main/dataset/retail\_insights\_dataset.csv
```

The file contains **5,000 order-line records** across 789 unique customers, 3 product categories (Office Supplies, Technology, Furniture), and 4 customer types (Consumer, Corporate, Home Office, Small Business), spanning orders shipped to Sydney, NSW and Melbourne, VIC.



Data Dictionary

Column	Description
Order No	Unique order-line identifier, e.g. 4293-1.
Order Date	Date the order was placed (format DD-MM-YYYY).
Customer Name	The customer who placed the order. The same name can appear on multiple rows — this dataset has real repeat customers (789 unique customers across 5,000 order lines), which is exactly what makes true RFM feature engineering possible this week.
Address / City / State	Delivery location. Every order ships to either Sydney, NSW or Melbourne, VIC.
Customer Type	Business segment of the customer: Consumer, Corporate, Home Office, or Small Business.
Account Manager	The staff member who manages this account.
Order Priority	Fulfilment urgency: Critical, High, Medium, Low, or Not Specified.
Product Name / Product Category / Product Container	What was purchased. Product Category is one of Office Supplies, Technology, or Furniture.
Ship Mode / Ship Date	How and when the order shipped.
Cost Price / Retail Price / Profit Margin	Unit-level cost, retail price, and profit margin, stored as currency strings (e.g. "\$156.50") — you will need to strip the "\$" and any thousands-separator commas before converting to float.
Order Quantity	Number of units in this order line.
Sub Total / Discount % / Discount \$ / Order Total / Shipping Cost / Total	The full pricing breakdown for the order line, also stored as currency/percent strings. Order Total (or Total) is the natural choice for your Monetary feature — briefly justify whichever one you pick.

2. Your Task 🎯

Suggested Problem Framing

(you may follow this OR propose your own approach, clearly justified)

Aggregate the 5,000 order-line records up to the customer level (group by Customer Name) and build an RFM-style feature set:

- **Recency** — days between each customer's most recent Order Date and a fixed reference date (e.g. the maximum Order Date in the dataset, plus one day).
- **Frequency** — number of distinct orders placed by that customer.
- **Monetary** — total spend by that customer (sum of Order Total or Total across all their orders).

Then add at least two more behavioral features of your own choosing — for example: average discount % received, average profit margin generated, dominant Product Category (encoded), dominant Customer Type, or the proportion of Critical/High-priority orders. This mirrors exactly how we built our customer-level feature set in class, just with real repeat-purchase data this time.

Choose ONE clustering approach — K-Means, DBSCAN, or a deep-learning-based approach (autoencoder + clustering) — and justify your choice using the Decision Guide criteria from class

(shape of the data, whether you know K in advance, dimensionality, need for outlier detection, etc.). You do not need to run all three like we did in the in-class notebook; picking one and doing it well, with a clear justification for why you picked it, is what's expected here.

Required Steps

(mirrors our in-class notebook)

1. EDA & cleaning — clean the currency columns (strip "\$" and thousands-separator commas, convert to float), parse Order Date and Ship Date, check for missing values (Address and Order Quantity each have 1 missing row), and plot the distribution of Order Total/Total, Customer Type, Product Category, and Order Priority.
2. Feature engineering — build the customer-level RFM feature set described above, plus your two extra behavioral features.
3. Feature scaling — apply StandardScaler to your feature set and explain, in your own words, why this step is not optional for K-Means/DBSCAN, exactly as covered in Section 5 of the in-class notebook.
4. Dimensionality reduction — apply PCA, report the explained variance of the first few components, and use a 2D PCA projection to visualize your data before and after clustering.
5. Choose your clustering configuration — if using K-Means (or GMM), select K using at least two of the four metrics from class (Elbow, Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Index); if using DBSCAN, tune eps and min_samples using a k-distance plot, exactly as demonstrated in Section 10.
6. Train your chosen clustering model on the scaled (and/or PCA-reduced) features.
7. Validate your result — report the Silhouette Score at minimum (Davies-Bouldin / Calinski-Harabasz if relevant); for DBSCAN, also report the number and percentage of points flagged as noise.
8. Business Interpretation (this is the most heavily-weighted step this week — see the Grading Emphasis box below) — profile every single cluster using real numbers (average Recency, Frequency, Monetary, dominant category, dominant customer type, discount behavior, etc.) and give each cluster a clear, memorable persona name.
9. Recommend at least one concrete, profit-maximizing action per cluster, and quantify the expected business impact using clearly labeled assumptions — exactly like the win-back campaign estimate we built together in class.
10. Conclusion — which cluster is the most valuable to the business, which is the most at-risk of churning, and if you only had budget for one campaign this quarter, which cluster would you target first, and why?

Grading Emphasis — Read This Carefully

Running the clustering algorithm correctly is necessary, but it is not sufficient to score well this week. Most of the marks for this assignment are earned in Steps 8–10 above: how clearly you can explain, in plain business language backed by real numbers from your own cluster profile table, what each cluster actually represents — and what specific, profit-driving action you would take for each one.

A submission with a slightly weaker clustering result but a crisp, well-argued, numbers-backed business explanation for every cluster will score higher than a submission with a technically perfect clustering pipeline and only generic, one-line cluster descriptions.

3. Bonus Points 🌟

- Use a clustering algorithm we did not build hands-on in class (e.g. Spectral Clustering, HDBSCAN, Mean Shift, or Gaussian Mixture Model if your main model was K-Means or DBSCAN) — +bonus
- Compare two different clustering approaches side by side (e.g. K-Means vs. DBSCAN) on the same feature set, and discuss which one is more useful for the business, not just which scores higher on Silhouette — +bonus
- Engineer a customer loyalty or churn-risk score from Recency + Frequency, and show how it changes the way you'd prioritize your clusters — +bonus
- Quantify the "what-if" profit impact of a proposed campaign for at least two different clusters (not just one), exactly like Section 15 of the in-class notebook — +bonus
- Build the autoencoder + clustering pipeline from Section 11.3 of the in-class notebook as an alternative to your main model, and compare its Silhouette Score and business usefulness against your primary approach — +bonus



Special Recognition Award

The student(s) who give the clearest, best-explained example this week — meaning every single cluster is described in plain business language with supporting numbers, and paired with a specific, well-justified profit-maximizing action — will receive +1% extra credit added to this week's assignment score.

Make sure your final cluster explanations and recommended actions are easy to find in your notebook's conclusion so they can be verified during grading.